DESIGN AND FABRICATION OF A 3D INTERLOCKING STRUCTURE

Using Fusion 360 and CO_2 Laser Cutting Technology

Material Thickness: 2 mm
Software Used: Autodesk Fusion 360, RDWorks V8
Manufacturing Process: Laser Cutting

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1. Abstract

This project presents the complete workflow involved in designing, developing, and fabricating a 3D interlocking circular structure using CAD modeling and laser cutting technology. The model was designed in Autodesk Fusion 360 and fabricated using a CO_2 laser cutting machine controlled through RDWorks V8 software.

The structure is made entirely from 2 mm thick plywood and assembled using a slot-fit mechanism without adhesives or mechanical fasteners. Special attention was given to kerf compensation, slot tolerance, symmetry alignment, and structural stability.

The project demonstrates practical application of digital fabrication, precision engineering, and material-efficient manufacturing.

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2. Introduction

Laser cutting is one of the most widely used digital fabrication technologies in modern manufacturing. It allows high-precision cutting of materials with minimal material waste.

This project focuses on:

- Transforming a 3D CAD model into flat 2D cut profiles
- Understanding material tolerance
- Implementing slot-based structural assembly
- Fabricating a functional 3D structure from flat sheet material

The final outcome is a geometrically symmetric interlocking structure that showcases structural stability through press-fit design principles.

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3. Design Concept and Structural Logic

The structure is composed of:

- Curved outer support panels
- Circular stabilizing discs
- Central perpendicular intersecting plates
- Slot-based interlocking joints

The design principle is based on:

Radial Symmetry

All curved panels are symmetrically arranged around the central axis to distribute load uniformly.

Orthogonal Intersections

Vertical plates intersect at 90° to provide central rigidity.

Slot-Fit Assembly

Each slot is designed to match the material thickness (2 mm), enabling friction-based assembly.

The final geometry achieves structural balance without screws, glue, or adhesives.

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4. CAD Modeling in Autodesk Fusion 360

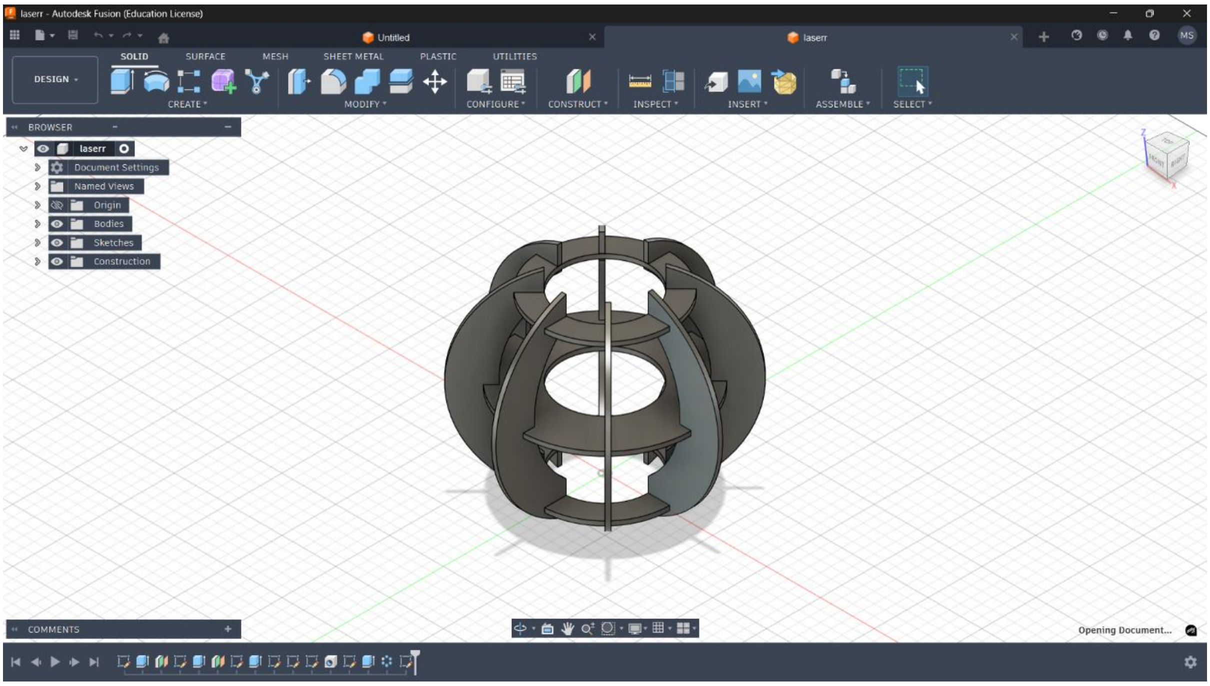
4.1 Sketch Development

The design process began with 2D sketching:

- Arcs used to create outer curved panels
- Circles used for internal support rings
- Rectangular slots added for interlocking
- All dimensions fully constrained

Special care was taken to:

- Maintain equal spacing
- Ensure symmetrical alignment
- Avoid overlapping geometry



Fusion 360 3D Model Screenshot
Caption: *Fully Assembled 3D Model Designed in Fusion 360*

4.2 Parametric Control

The model was created using parametric design principles:

- Material thickness parameter = 2 mm
- Slot width linked to thickness parameter
- Easy modification capability

This ensures scalability for other material thicknesses (e.g., 3 mm, 4 mm).

4.3 3D Assembly Validation

After extruding all parts to 2 mm thickness:

- Components were aligned using joint tools
- Interference check was performed
- Fit clearance verified

The assembled model confirms proper alignment and slot engagement.

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5. DXF Preparation and Export

To prepare for fabrication:

1. All components were flattened into 2D profiles
2. Duplicate sketch lines were removed
3. Profiles were confirmed as closed loops
4. Units verified in millimeters
5. DXF file exported

Accuracy at this stage is critical to avoid machine errors.

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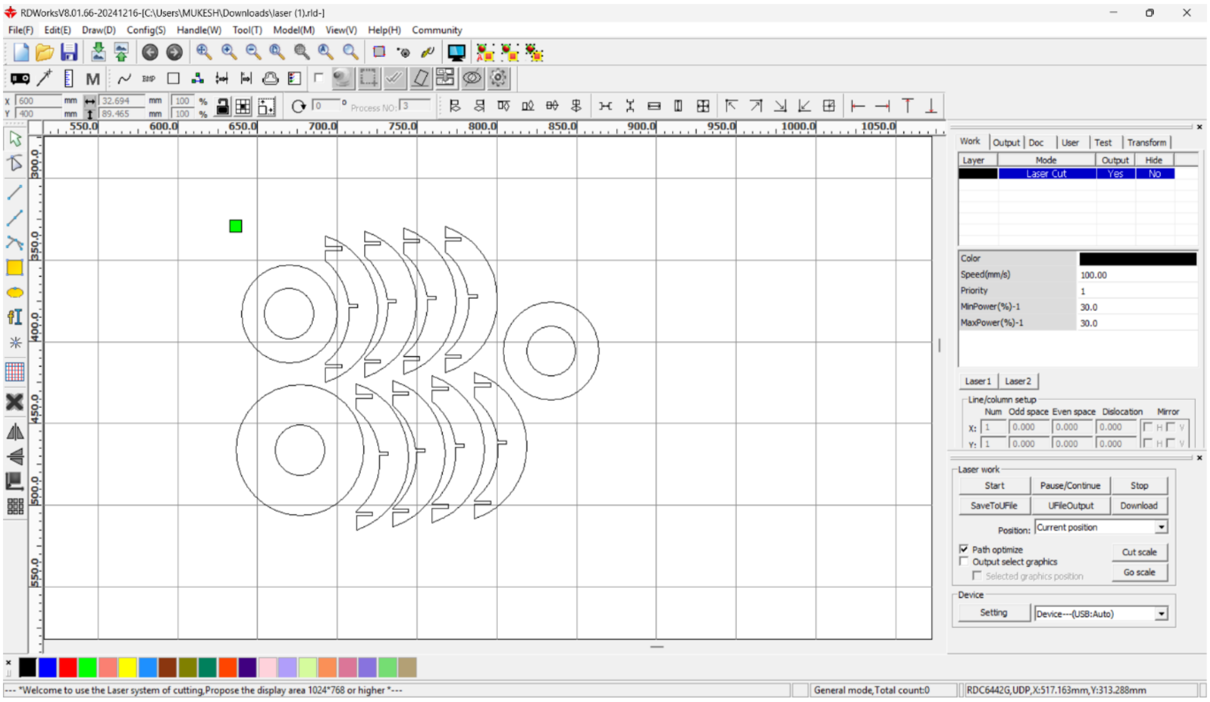
6. Laser Cutting Process Using RDWorks V8



RDWorks V8 Cutting Layout Screenshot
Caption: *Laser Cutting Layout Prepared in RDWorks V8*

6.1 Layout Optimization

The cutting layout was arranged to:

- Minimize material waste
- Reduce cutting time

- Maintain safe spacing between parts
- Avoid thermal overlapping

Manual nesting was used to optimize sheet utilization.

6.2 Machine Parameters

Material: 2 mm Plywood
Machine Type: CO_2 Laser Cutting Machine

Typical Settings Used:

- Speed: 80–100 mm/s
- Power: 25–35%
- Air Assist: Enabled
- Cutting Pass: Single

These parameters ensured:

- Clean cut edges
- Minimal burn marks
- Full penetration
- Dimensional accuracy

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7. Fabrication Procedure

1. Wooden sheet placed on honeycomb bed
2. Laser focus calibrated
3. Frame test performed
4. Cutting executed
5. Components removed carefully
6. Edges cleaned
7. Slot fitting tested

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8. Final Assembled Model

Final Assembled Wooden Model Photo
Caption: *Final Fabricated and Assembled 3D Interlocking Structure*

The final model demonstrates:

- Excellent structural rigidity
- Perfect slot alignment
- Uniform symmetry
- Clean laser-cut finish

The press-fit joints provide sufficient friction to maintain stability.

9. Engineering Analysis

9.1 Kerf Compensation

Laser kerf (material removed by laser beam) typically ranges between 0.08 – 0.15 mm.

If ignored, it may cause:

- Loose fitting
- Dimensional mismatch

Therefore, slot width was adjusted considering kerf tolerance.

9.2 Structural Stability

The structure achieves stability through:

- Radial load distribution
- Intersecting vertical supports
- Balanced geometry

The circular discs act as load distributors while curved panels provide outer reinforcement.

9.3 Material Behavior

2 mm plywood offers:

- Lightweight characteristics
- Sufficient stiffness
- Easy machinability
- Smooth laser cut edges

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10. Challenges Faced

- Slot tightness adjustment
- Minor burn marks during first trial
- Alignment correction during assembly
- Parameter optimization in RDWorks

These were resolved through testing and iteration.

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11. Learning Outcomes

Through this project, I developed hands-on expertise in:

- Digital product development
- CAD modeling and parametric design
- Laser cutting workflow
- Tolerance and kerf management
- Machine parameter optimization
- Fabrication-based problem solving

This project strengthened my understanding of precision manufacturing and digital prototyping technologies.

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12. Personal Contribution

I independently:

- Designed the complete model in Fusion 360
- Prepared DXF files

- Configured RDWorks parameters
- Operated the laser cutting machine
- Performed final assembly

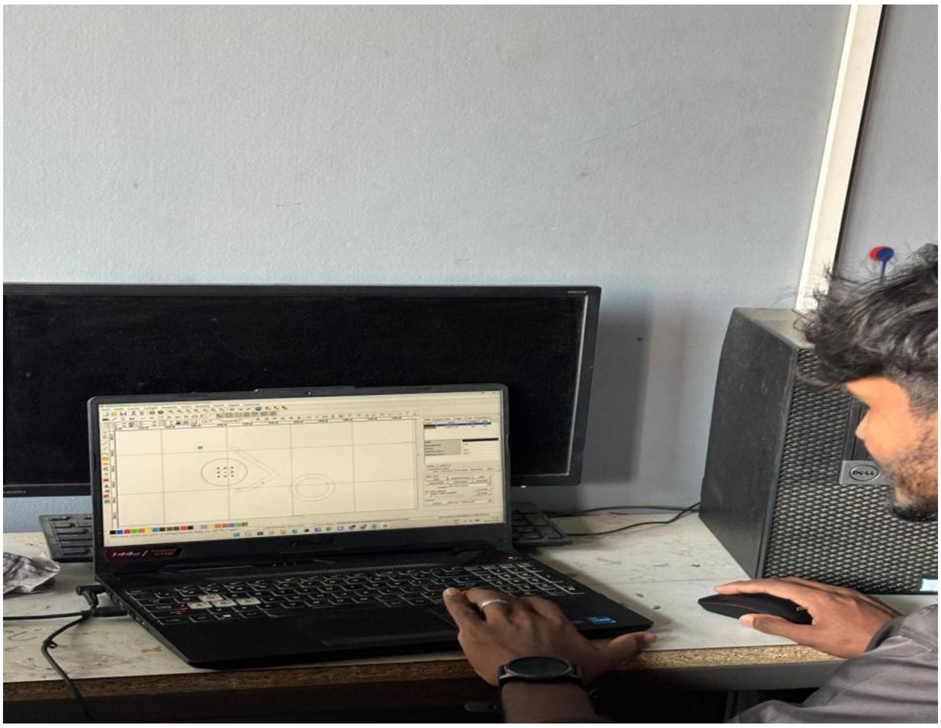

Photo of Me Operating the Laser Cutting Machine
Caption: *Hands-on Fabrication Process – Operating Laser Cutting Machine*

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13. Conclusion

The project successfully demonstrates the integration of CAD modeling and laser cutting fabrication to produce a structurally stable 3D interlocking structure from 2 mm sheet material.

The slot-based assembly technique eliminates the need for fasteners while maintaining mechanical strength. The project highlights the importance of precision design, tolerance management, and fabrication workflow optimization in modern digital manufacturing.
